

BENJAMIN BONNELL

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EDUCATION

Virginia Tech | Alexandria, VA | August 2025 - May 2027

M.Eng, **Computer Science**: Software Development & Applications

James Madison University (JMU) | Harrisonburg, VA | August 2022 - May 2026

B.S. **Computer Science**, Minor in ISAT: **Industrial & Manufacturing Systems**

Cum Laude - **3.51/4.00 GPA**

WORK EXPERIENCE

Project Intern | E. E. Reed East Coast | Fredericksburg, VA | May 2026 - Present

- Communicate with subcontracting companies to ensure synchronization between field work and plans through online messaging & on-site verbal conversations.
- Work with senior staff to analyze budgeting & planning and identify cost-saving measures

Industrial & Manufacturing TA | JMU CISE | Harrisonburg, VA | March 2025 - May 2026

- Assisted groups of 20-30 students with Fusion360 CAD and CAM software
- Operated and demonstrated manufacturing equipment such as metal casting, 3D printers, metal shaping machines, material testing machines, and CNC machinery.
- Oversaw open manufacturing laboratory, was responsible for maintaining lab safety, assisting students with equipment setup, troubleshooting, and project support.

Technology Coordinator | iD Tech | Villanova, PA | May 2025 - August 2025

- Led instruction for groups of students, teaching foundational robotics and software concepts.
- Reduced equipment downtime by several days by designing and 3D printing replacement parts for damaged equipment.

Student IT Assistant | JMU CSM | Harrisonburg, VA | January 2024 - May 2025

- Coordinate with staff and team to effectively deploy and maintain workstations and servers across JMU Campus.
- Manage robust inventory system for JMU's East Campus academic buildings/offices.

Intern | JMU X-Labs | Harrisonburg, VA | August 2022 - December 2022

- Responsible for operating technical software and using fabrication equipment such as 3D printers, Laser Cutters, etc.
- Work with team members to provide product design & development assistance to clients.

LEADERSHIP & EXTRACURRICULAR EXPERIENCE

Web Designer | WXJM Radio | Harrisonburg, VA | June 2023 - May 2026

- Led a multidisciplinary team to build & maintain the official [WXJM.org](https://www.wxjm.org) website.
- Ensured continuous operation of all web-related systems, promptly addressing & resolving technical issues whenever they may arise.
- Complete all responsibilities in a fully remote and asynchronous environment.

Professional Development Chair | Theta Tau | JMU | August 2025 - December 2025

- Worked with a team to organize professional development events, including resume workshops & hosting guest speaker panels.
- Participated in STEM-related community service events in the local area.

PROJECT EXPERIENCE

HazBot Autonomous Robot | Spring 2026

- Designed and built an arduino-based robot to complete an obstacle course, with a variety of sensors
- Wrote arduino code to utilize various sensors and motors with different libraries
- Tuned sensors and motors for 100% accuracy on course.
- Designed and 3D Printed a chassis for the components and wiring, created CAD assembly and drawings.

WebGL Video Game | Fall 2025

- Created a Frogger-style web-based platformer game from scratch.
- Wrote WebGL engine from scratch, implemented library in TypeScript to draw complex mesh models to the screen, with support for shaders.
- Modeled and textured all 3D game assets in Blender, with animations built from motion-capture data.

RimpleX Complex Number Calculator | Spring 2025

- Built a complex number calculator with Java & Swing GUI, in a simulated software engineering environment.
- Worked in a team of engineers under SCRUM framework, conducted successful meetings and negotiations with the project manager.

Pomodoro Timer | Summer 2024

- Built a functional study timer app with HTML/CSS/JavaScript
- Used SCRUM process framework to delegate tasks, review progress, and handle sprints.
- Designed UI from scratch in HTML and CSS, made mascot GIFs in Blender.

SKILLS

SCRUM, Git, Python, Java, JUnit, JavaScript, TypeScript, C, C#, HTML, React, Linux, Code Review, Project Presentations, WebGL, Three.js, CAD, Object Oriented Programming, UI/UX, AI-Assisted Development

Relevant Coursework: Software Engineering, Advanced Programming, Computer Systems I & II, Full Stack Application Development, Data Structures & Algorithms, Applied Algorithms, Computer Graphics, Issues in Modern Production Systems, Production Systems Automation, Programming Languages